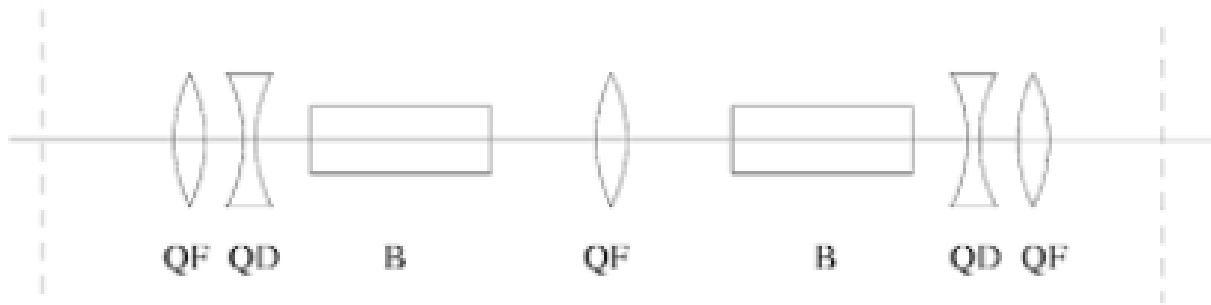


Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

Problem 1. NSLS II Longitudinal Parameters (Electrons) 50 pts.

NSLS II adopts a double bent achromat, DBA, lattice.



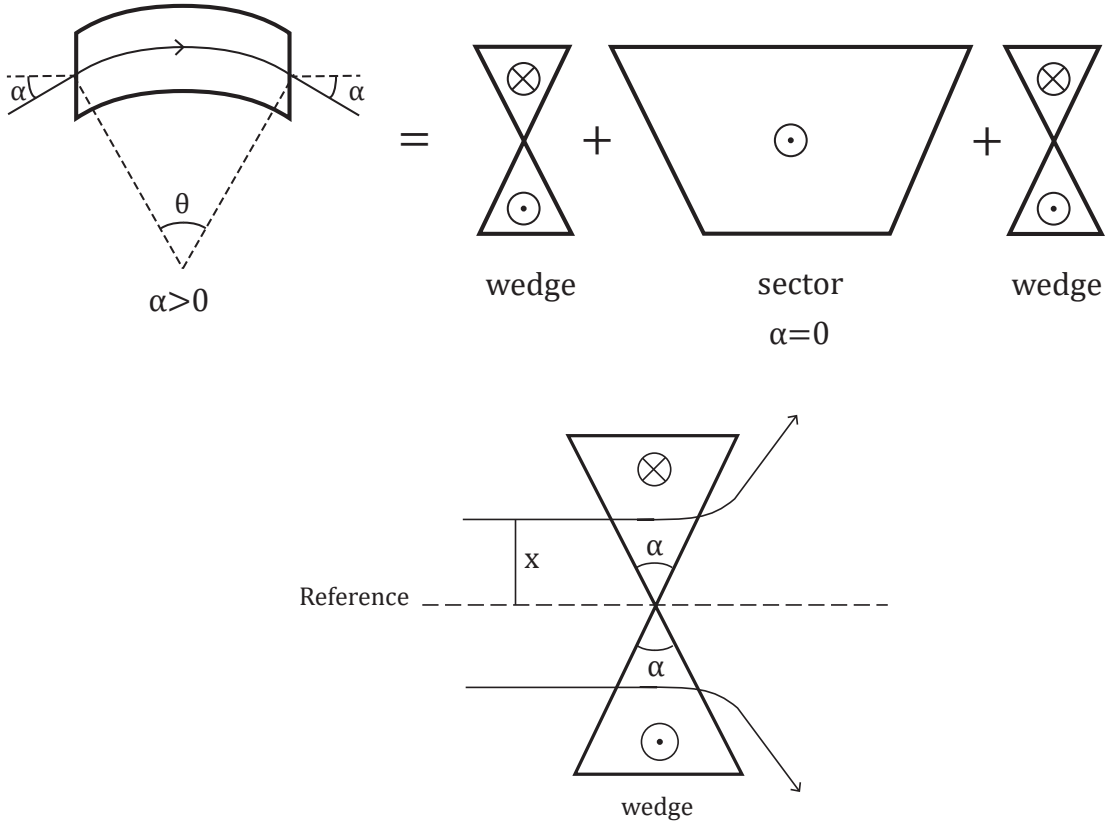
Parameters	Values
Energy [GeV]	3.0
Circumference [m]	780
Number of dipoles	60
Dipole field [T]	0.4
Beam current [A]	0.5
RF frequency [MHz]	499.68
Harmonic number	1320

From the design parameters of NSLS II shown in the table above, calculate the following:

- Calculate the axial length of the dipoles assuming all dipoles are the same. (5pts)
- In the DBA lattice, the dispersion, D , and dispersion slope, D' , are zero at one end of the dipoles and non-zero at the other end. Find the dispersion inside the dipole magnet (10 pts).
- Calculate the energy loss due to the dipole field (10 pts).
- If the synchronous accelerating phase of the RF cavity is $\pi/6$, what is the minimum RF voltage required? How much power is required? (5pts)
- The actual RF voltage is about 3 MV. Using this value, find the longitudinal tune of NSLC II (5 pts).
- What is the critical radiation frequency of the dipole radiation? (5pts)
- Find the partition number, $\overline{D}$, due to the synchrotron

- h Find the longitudinal damping rate, α_E , and compare with the period of longitudinal oscillations (5 pts).

Problem 2. Dipole Edge Corrections - Horizontal 20 pts.



- a) 15 pts: Horizontal correction:

A dipole with entry and exit angles α (defined relative to reference orbit, see figure above) can be divided into three sections: a sector dipole in the center with a wedge magnet on both ends. The sector dipole has the transfer matrix derived in class.

For a short sector, argue from the Lorentz force equation that the particle experiences the following corrections for the orbit displacement x through the wedge:

$$\Delta x' = \frac{B_y(0)}{[B\rho]} \tan(\alpha) x$$

$$\Delta x = 0$$

to show that:

$$\mathbf{M}_{\text{wedge}} = \begin{bmatrix} 1 & 0 \\ \frac{\tan(\alpha)}{\rho} & 1 \end{bmatrix}; \quad \frac{1}{\rho} = \frac{B_y(0)}{[B\rho]}$$

This “kick” correction will be applied entering and exiting the magnet.

- b) 5 pts: Show that the x -transfer matrix of the full dipole can be expressed as:

$$\mathbf{M}_x = \mathbf{M}_{\text{wedge}} \cdot \mathbf{M}_{\text{sector}} \cdot \mathbf{M}_{\text{wedge}} = \begin{bmatrix} \frac{\cos(\theta-\alpha)}{\cos \alpha} & \rho \sin(\theta) \\ -\frac{\sin(\theta-2\alpha)}{\rho \cos^2(\alpha)} & \frac{\cos(\theta-\alpha)}{\cos \alpha} \end{bmatrix}$$

Hint:

$$\begin{aligned} \cos(\theta) + \tan(\alpha) \sin(\theta) &= \frac{\cos(\theta - \alpha)}{\cos(\alpha)} \\ \sin(\theta) - \tan(\alpha) \cos(\theta) &= \frac{\sin(\theta - \alpha)}{\cos(\alpha)} \end{aligned}$$

Problem 3. Dispersion In Extraction Line 20 pts.

The dispersion function in a period ring satisfies:

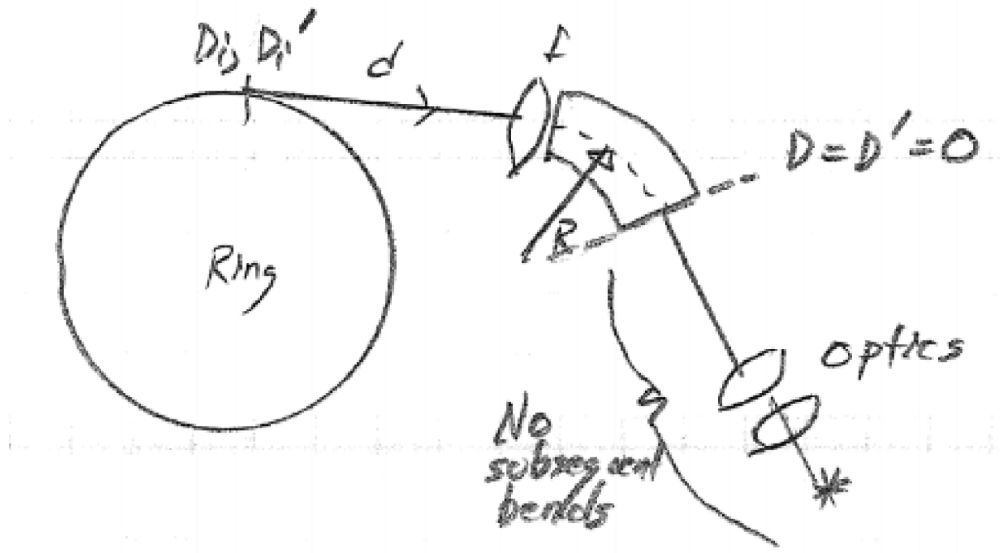
$$D''(s) + K(s)D(s) = \frac{1}{\rho(s)}$$

$\rho(s)$ = bend radius, $K(s)$ = focusing function

$$D(s + L_p) = D(s), \rho(s + L_p) = \rho(s), K(s + L_p) = K(s)$$

L_p = lattice period

- a) 5 pts: Argue that the solution for D is unique. This implies that there is a unique closed orbit $x = \delta D$ for every value of off-momentum δ . (Hint: let D_1 and D_2 be two independent solutions and look for a contradiction.)
- b) 15 pts: A particle is kicked out of a ring with dispersion $D = D_i$ and $D' = D'_i$ just after the kick. The particle is then transported through an extraction line with a drift length d , a thin lens focusing kick with focal length f , and a sector bend of radius $\rho = R$ and length ℓ , and finally through an unspecified series of optics to the target. (See diagram below).



Find a constraint on the lattice parameters d , f , R , and ℓ that can be enforced to ensure that $D = D' = 0$ after the bending magnet (as indicated) to have zero dispersion in the straight transport and focusing line to the target. (Assume the elements can all be treated in the thin approximation.)